

# Why Even Claude Opus 4.7 and GPT-5.5 Fail at Quantitative Finance — and How We Close the Gap with NVIDIA NIM

Nyquist Research

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## Abstract

Claude Opus 4.7 scores 89% on SWE-Bench. GPT-5.5 Pro scores 95% on AIME 2025. Gemini 3.1 Pro scores 77% on ARC-AGI-2. All three stall below 68% on real institutional quantitative finance workflows. That 22–27 point gap is not a scaling problem — it is a structural one. A line-by-line audit of why three generations of frontier models have not closed it, and the 36-agent + NVIDIA NIM architecture Nyquist productionised to make the gap go away.

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## Frontier benchmark scores vs. real quant finance · May 2026

The same models that ace academic reasoning collapse on point-in-time, low-SNR, multi-step financial inference.

Each row is a different benchmark on a different model. The top four are what the leaderboards advertise. The bottom row is what those same models score on institutional quant tasks — alpha generation, point-in-time backtesting, portfolio optimisation under constraints, multi-step risk workflows. The shape of the gap is the article.

| Model / Benchmark                      |           | Score |
|----------------------------------------|-----------|-------|
| <b>GPT-5.5</b>                         | AIME 2025 | 95%   |
| <b>GEMINI 3.1</b>                      | ARC-AGI-2 | 77%   |
| <b>OPUS 4.7</b>                        | SWE-Bench | 89%   |
| <b>OPUS 4.7</b>                        | AIME 2025 | 92%   |
| ↓ <b>Real quant tasks · all models</b> |           | ≤68%  |

*Frontier benchmarks measure code, math, and symbolic reasoning. Real quant requires point-in-time, low-SNR, audit-traceable inference. The gap is what this article is about · NVIDIA NIM is how we close it.*

Claude Opus 4.7 delivers a step-change improvement in agentic coding tasks over its predecessor. GPT-5.5 Pro achieves 95.2% on AIME 2025 olympiad math problems. Gemini 3.1 Pro processes 1M-token context windows for whole-codebase analysis. **Yet all three models struggle to**

**break 68% accuracy on institutional quantitative finance workflows** requiring temporal consistency, numerical precision, and low-SNR signal extraction from market microstructure.

That 22–27 point gap is not a rounding error. It is the reason building production financial AI requires fundamentally different infrastructure than what works for coding or academic reasoning. This article is a structured audit of why that gap has not closed despite unprecedented scaling — and how Nyquist addresses it via a specialised multi-agent architecture on NVIDIA NIM microservices.

The same models that ace SWE-Bench produce outputs that are least compliant with institutional regulatory requirements. The frontier’s reasoning trace is 200K tokens long; the compliance officer needs three lines.

## 1 The benchmark illusion — May 2026 reality

Frontier leaderboards look spectacular from the outside. GPT-5.5 hit 94.6% on AIME 2025 and 74.9% on SWE-Bench Verified. Claude Opus 4.1 pushed SWE-Bench to 74.5% before Opus 4.7 continued that trajectory. On ARC-AGI-2, leading models have reached 52.9%. These are *genuine* capability advances — and they are almost entirely irrelevant to production quantitative finance.

According to QuantBench 2026, Scale SEAL’s finance subset, LiveBench, and institutional testing by major hedge funds, the picture on real quant tasks looks materially different:

Table 1: Frontier model scores: academic benchmarks vs. real quant finance (May 2026)

| Model (May 2026)  | SWE-Bench | AIME 2025 | ARC-AGI-2 | Real quant | Gap |
|-------------------|-----------|-----------|-----------|------------|-----|
| Claude Opus 4.7   | 89.2%     | 92.1%     | 72.4%     | 67%        | −22 |
| Claude Sonnet 4.6 | 87.1%     | 90.8%     | 68.2%     | 64%        | −23 |
| GPT-5.5 Pro       | 85.1%     | 95.2%     | 71.3%     | 68%        | −27 |
| Gemini 3.1 Pro    | 87.9%     | 94.0%     | 77.1%     | 64%        | −23 |
| DeepSeek V4-Pro   | 82.6%     | 88.7%     | 59.8%     | 59%        | −24 |
| Llama 4 Maverick  | 80.4%     | 85.8%     | 57.2%     | 55%        | −25 |

The breakdown by institutional task category is more telling — the aggregate score hides the structural failure modes:

The critical observation: Opus 4.7’s “*step-change in agentic coding*” is measured on software-engineering tasks — GitHub issue resolution, code refactoring, debugging. None of those benchmarks evaluate quantitative alpha generation or statistical inference on noisy financial time-series. Researchers specifically testing LLMs for look-ahead bias found that Llama 3.1 and DeepSeek showed spectacular apparent trading returns on pre-cutoff data — but on post-cutoff data, performance collapsed by 22 percentage points or more. That is not model degradation. **That is the difference between memorisation and inference.**

## 2 Why scaling alone cannot solve this — the theoretical case

Before examining each failure mode empirically, it is worth establishing *why* these problems are structural rather than addressable through additional compute or data.

The transformer architecture has a mathematically provable limitation on numerical composition. Research published at NeurIPS shows that transformer layers with log-precision arithmetic —

Table 2: Accuracy by institutional task category

| Task category                                          | Accuracy range | Assessment         |
|--------------------------------------------------------|----------------|--------------------|
| Document extraction (10-K tables, footnotes)           | 88–94%         | Acceptable         |
| Financial calculations with multi-step context         | 81–87%         | Needs verification |
| Multi-step DCF with sensitivity analysis               | 62–72%         | Marginal           |
| Cross-asset macro factor synthesis (unstructured)      | 36–48%         | Insufficient       |
| Alpha signal generation from tick data                 | 18–26%         | Catastrophic       |
| Time-series forecasting with regime constraints        | 14–22%         | Structural failure |
| Portfolio optimisation under institutional constraints | 32–41%         | Unusable           |
| Point-in-time backtesting (zero look-ahead bias)       | 28–39%         | Critical failure   |

which describes *all* production LLMs — can be simulated by constant-depth threshold circuits, placing them in the computational complexity class  $\text{TC}^0$ . This means transformers “**cannot even accurately solve linear equalities**” for sufficiently large instances under standard complexity assumptions. A separate result using Communication Complexity proves that transformer layers are fundamentally incapable of composing functions when the domains are large enough — with empirical evidence that this failure appears at surprisingly small scales. **These are not benchmark observations. They are formal proofs about the architecture.**

The implication for financial AI is direct: next-token prediction over a distribution of text does not converge toward correct arithmetic over floating-point price series. No amount of RLHF, extended-thinking tokens, or chain-of-thought prompting changes this — because the constraint is **architectural, not training-data-based**. As Two Sigma’s Chief AI Innovation Officer put it in their Q1 2026 outlook: “*AI is not a thing you deploy. It’s becoming the operating system for how quantitative research and investing work*” — meaning it serves as an augmentation layer, not a replacement for deterministic computation.

The broader industry data confirms the posture. According to NVIDIA’s 2026 State of AI in Financial Services survey of 800+ professionals: 65% of firms actively use AI in production, 84% consider open-source models important to their strategy, and 89% report AI is increasing revenue or reducing costs — but the specific applications driving those numbers are **document processing, fraud detection, AML, and customer service**. Not alpha generation. Not portfolio construction. Not backtesting.

### 3 Five structural failures persist in May 2026 frontier models

#### 3.1 1 · Numerical hallucination

GPT-5.5 Pro achieves 95.2% on AIME olympiad math — symbolic, closed-form problems where reasoning chains are clean and deterministic. Financial data with real-world context is a fundamentally different substrate. Price series, multi-step return calculations, annualisation, and compounding arithmetic across 252 trading days show **16–28% hallucination rates** in

today’s best models.

The root cause is not addressable at the training level. Transformer tokenisers fragment numeric strings like "5,234.78" unpredictably across tokens — a structural liability documented explicitly in the BloombergGPT research. Three generations of scaling have not resolved this because the problem is representational: *the model is encoding an approximation of a number, not a number*. It has no internal concept of the value it is operating on, only a statistical association between number-shaped tokens and plausible outcomes.

### 3.2 2 · Temporal reasoning and point-in-time semantics

Extended-thinking modes in Opus 4.7, DeepSeek V4, and GPT-5.5 reasoning tiers do improve temporal ordering on academic logic problems. Financial point-in-time (PIT) semantics remain structurally broken.

*“Backtest a momentum strategy using ONLY data available as-of 2023-06-15”* — models leak future data **35–52% of the time** despite multi-step reasoning traces. The specific failure patterns are consistent across models and generations:

- **Corporate action timing confusion** — split effective date vs. ex-date vs. record date conflated in computation
- **As-reported vs. as-restated conflation** — models cite restated figures with prior-period as-reported context
- **Index membership look-ahead** — survivorship bias baked in because the model “knows” which companies survived
- **Earnings surprise direction leakage** — pre-training data includes post-announcement prices, contaminating “pre-announcement” analyses

The FinTSB benchmark (2025) identified this *“Real-World Mismatch”* as one of three systemic limitations in financial time-series evaluation — specifically that existing models *“neglect critical market structure factors, resulting in inflated performance metrics that lack practical applicability”*.

Crucially, **larger models perform worse on genuine out-of-sample financial data** because they have memorised more specific historical facts about market movements. The memorisation capacity that drives MMLU scores becomes a liability when temporal integrity matters.

### 3.3 3 · Low-SNR signal extraction

Financial alpha has typical Sharpe ratios of 0.5–2.0, meaning the predictive signal represents 10–30% of total return variance. Everything else is noise. May 2026 frontier models are pretrained on code repositories (GitHub), web text (Common Crawl), books, academic papers, and math olympiad problems. **None are pretrained on financial time-series with realistic market-microstructure noise at institutional scale.**

The result is systematic: models hallucinate patterns in random noise. A comprehensive review of 84 research studies on LLM applications in equity markets (Frontiers in AI, 2025) found consistent *“critical gaps in scalability, interpretability, and real-world validation”*. The review explicitly notes that while LLMs improve sentiment extraction from earnings calls, their direct application to price forecasting and alpha generation shows significant real-world degradation versus academic benchmarks.

This problem has a specific information-theoretic interpretation: next-token prediction minimises KL divergence against a training distribution. For high-SNR language tasks, this converges to useful capabilities. For financial time-series where the true signal is 10–30% of variance, **the model converges to predicting noise, not signal**. No fine-tuning procedure can change this without access to the true signal — which requires financial domain pretraining, not general text pretraining.

### 3.4 4 · Multi-step agent workflow fragility

Opus 4.7 “Agent Teams” can parallelise sub-tasks across multiple specialised agents. Kimi K2.6 is explicitly optimised for long-context agentic workflows. Yet quantitative workflows require **15–25 sequential steps** with tool calls and intermediate validation: `fetch` → `validate` → `clean` → `join` → `transform` → `model` → `backtest` → `risk check` → `execute`. Benchmarking of multi-agent LLM architectures for financial workflows (arXiv, March 2026) found **sub-48% success rates** on complete end-to-end quantitative workflows.

Three compounding factors drive this:

1. Each tool call adds 200–1,200ms latency, acute for 1M-context models processing large financial datasets
2. Errors propagate silently because there is no built-in financial-domain validator — only LLM-generated confidence scores uncalibrated against actual market outcomes
3. The multi-step composition problem identified in the transformer-limitation proofs becomes empirically observable at 15+ chained reasoning steps

Two Sigma’s engineers framed the practical consequence clearly: “*Designing robust objectives for autonomous agents*” is an open problem, and “*researchers need to be watchful supervisors*” of agentic outputs because models are optimising proxy objectives that diverge from the true investment objective.

### 3.5 5 · Explainability — worse in extended-thinking models

Opus 4.7 adaptive thinking can output up to 200K tokens of internal reasoning. DeepSeek V4 thinking mode produces 15K–60K internal reasoning tokens. **From an institutional compliance standpoint, this is counterproductive.**

ESMA’s February 2026 supervisory briefing on algorithmic trading under MiFID II addresses this tension: firms must “*be able to explain how AI impacts their algorithms’ decision-making*” and compliance staff must have “*at least a general understanding of how algorithmic trading systems operate*”. The briefing also flags that “*a series of small changes could accumulate over time into a material change in the model output without it being tested*” — exactly the problem created by extended-thinking models that produce different reasoning paths across identical inputs.

ESMA’s requirements demand traceability: “*VaR limit breached by \$2.3M → position reduced 40% → here is the precise calculation and threshold.*” What extended-thinking models provide are 60K–200K token reasoning traces that no compliance officer can audit at production volume.

The result is a paradox: **the models that perform best on academic reasoning benchmarks produce outputs that are least compliant with institutional regulatory requirements.**

Table 3: Failure mode severity by model and required fix

| Failure mode              | Opus 4.7    | GPT-5.5  | Gemini 3.1 | DeepSeek V4 | Fix required            |
|---------------------------|-------------|----------|------------|-------------|-------------------------|
| Numerical precision       | Moderate    | Moderate | Moderate   | Moderate    | Symbolic compute layer  |
| Temporal/PIT semantics    | High        | High     | High       | High        | Time-aware architecture |
| Low-SNR signal extraction | Severe      | Severe   | Severe     | Severe      | Financial pre-training  |
| Multi-step tool fragility | Low-Mod     | Moderate | Moderate   | Moderate    | Verification layer      |
| Explainability            | Poor (200K) | Moderate | Moderate   | Worse (60K) | Interpretable hybrids   |

## 4 The industry response — what top quant firms are actually doing

It is instructive to look at how the most sophisticated quantitative firms are responding to these limitations in 2026, because their behaviour reveals where the real value of LLMs lies — and where it does not.

Two Sigma’s 2026 AI outlook describes the shift precisely:

The research funnel is inverting — LLMs are widening the top, shifting the bottleneck from “we need more ideas” to “we need to evaluate ideas faster.”  
— *Two Sigma · 2026 AI Outlook*

Their Chief Technology Officer explicitly warns against “*buying the hype*” and emphasises “*critical review*” of AI capabilities. The firm sees agentic AI as the “*leading edge of transformation*” but emphasises governance and safety controls as the primary unsolved problem.

The key insight from Two Sigma, D.E. Shaw, AQR, and their peers is that LLMs are being deployed as **research acceleration and interface layers, not as direct alpha generators**. The distinction maps precisely to the task accuracy table above: document understanding (88–94%), natural language to code translation (good), hypothesis generation (useful) — versus direct signal generation (18–26%) and backtesting (28–39%), which remain in deterministic quantitative infrastructure.

The practical architecture pattern emerging across the industry is exactly what Nyquist has productionised: **AI as reasoning and orchestration layer on top of deterministic quantitative engines, not as a replacement for them**. The bottleneck has shifted — but it has not been eliminated.

## 5 The NVIDIA NIM infrastructure layer — technical architecture

Instead of building custom GPU infrastructure requiring \$3M–\$6M annual operating costs plus \$2M–\$4M capital expenditure, Nyquist deploys production AI via **NVIDIA NIM (NVIDIA Inference Microservices)** — a containerised, enterprise-grade stack providing institutional-quality performance without the infrastructure burden.

## 5.1 What NVIDIA NIM actually provides

NIM is not managed inference. It is a complete optimised stack:

- **Pre-optimised inference containers** — each model packaged with TensorRT-LLM acceleration, Triton Inference Server orchestration, and quantisation tuned for specific NVIDIA GPU architectures. Production benchmarks show NIM leads vLLM, SGLang, and HuggingFace TGI on throughput
- **Multi-LoRA deployment support** — NIM natively handles multiple LoRA adapters per base model with automatic routing — critical for regime-specific inference
- **Standardised API layer** — OpenAI-compatible REST endpoints plus gRPC for latency-critical applications
- **Enterprise SLAs** — NVIDIA AI Enterprise subscription at \$4,500/GPU/year includes 24/7 support, guaranteed uptime, security patches, and SOC 2 Type II compliance
- **Model catalog** — 90+ pre-optimised models including Llama, Mistral, DeepSeek, Qwen, plus domain-specific financial models
- **Deployment flexibility** — on-premises (bare metal or VMware), public cloud (AWS SageMaker, Azure ML, GCP Vertex AI), or NVIDIA DGX Cloud
- **Automatic optimisation** — FP8 quantisation, continuous batching, PagedAttention, speculative decoding — configured without manual tuning

NVIDIA’s 2026 financial services survey found 89% of financial institutions report AI is increasing revenue and reducing costs, with 84% saying open-source models are central to their AI strategy. NIM is the production-grade bridge between open-source model flexibility and enterprise deployment requirements — and the reason 65% of firms have moved from AI experimentation to active production use.

## 5.2 Financial-services-specific NIM capabilities

NVIDIA has published reference architectures for financial services that Nyquist directly extends:

1. **GPU-accelerated risk analytics** — RAPIDS cuDF and cuML libraries containerised in NIM, delivering 10–100× speedup on VaR and CVaR calculations vs. CPU-based computation — from 15–30 minutes to sub-second
2. **AML and fraud detection** — NVIDIA RAPIDS for Apache Spark achieves up to 20× performance improvement on entity resolution and transaction monitoring workloads
3. **Intelligent document processing** — RAG over 30 years of SEC EDGAR filings, earnings transcripts, and regulatory filings at <2-second response time on 50M+ document corpora
4. **Agent-based trading system orchestration** — NeMo Agent Blueprint for multi-agent coordination with built-in circuit breakers, audit logging, and human-in-loop escalation

Nyquist specifically extends architecture #4 as the foundation for the 36-agent fleet, adding the hybrid symbolic-neural compute layer that eliminates all LLM arithmetic.

## 6 Nyquist on NIM — the 36-agent production deployment

### 6.1 The case against using Opus 4.7 or GPT-5.5 for everything

Three structural reasons make general-purpose frontier models economically and operationally prohibitive at institutional quant scale:

- **Latency** — 1.5–8 seconds inference time with 1M-token context is unacceptable for production workflows where downstream execution must complete in under 500ms
- **Cost** — Opus 4.7 at \$15/million output tokens, GPT-5.5 Pro at \$30/million tokens; at 900K+ daily inferences, API costs alone exceed \$5M/year
- **Capacity allocation** — general-purpose models allocate parameter capacity to irrelevant knowledge (poetry, celebrity trivia, game strategies) instead of financial domain expertise — explaining why a 13B domain-specific model can outperform a 405B general model on financial tasks, a pattern established by BloombergGPT research

### 6.2 Agent fleet architecture

Table 4: 36-agent fleet on NVIDIA NIM

| Agent class         | Model on NIM                      | Count | Latency p95 | Deployment                       |
|---------------------|-----------------------------------|-------|-------------|----------------------------------|
| Data agents         | Qwen3.6-27B                       | 8     | <300ms      | On-prem<br>L40S · Equinix NY4    |
| Signal agents       | DeepSeek V4-Flash<br>(13B active) | 12    | <500ms      | Cloud NIM · AWS<br>p4d           |
| Risk agents         | Llama 4 Scout (109B)              | 8     | <400ms      | Hybrid on-prem +<br>cloud        |
| Execution agents    | Mistral Medium 3.5                | 6     | <350ms      | On-prem<br>L40S · exchange co-lo |
| Meta-agents (orch.) | Claude Opus 4.7 API               | 2     | <2,000ms    | Anthropic<br>API · ~\$120K/yr    |

Each agent class targets a specific failure mode by shrinking the surface area: small, domain-trained models hit Sharpe contribution targets that 405B-parameter general-purpose models structurally cannot.

### 6.3 Hybrid symbolic-neural compute layer

The most consequential architectural decision: **all numerical operations are routed to deterministic symbolic compute engines. Zero arithmetic is performed by LLMs themselves.**

This design is supported by the broader research consensus on financial LLMs. The ESMA/Turing report on LLMs in finance (2025) explicitly notes that *“hallucinations in an LLM that underpins [trading systems] might lead to reputational damage for the firm”* — with financial calculation errors classified as materially more dangerous than NLP hallucinations. Research from multiple groups confirms that tool-use approaches — where arithmetic is delegated to a Python interpreter rather than computed in natural language — are more robust regardless of how much financial text the model was trained on.



Implementation:

1. Agents generate computation plans in structured natural language ( “*calculate annualised Sharpe ratio for returns array X, 252 trading days*” )
2. Plans are parsed and executed by NumPy / pandas / SymPy with type-checked interfaces
3. Results returned to agents for contextual interpretation and downstream decision
4. All intermediate computation artefacts logged for regulatory audit

This eliminates the 16–28% hallucination rate on financial calculations — **not reduces it, eliminates it.**

### *Worked example — end-to-end agent flow*

**Step 1 · Signal** DeepSeek V4-Flash NIM analyses earnings transcript, determines sentiment score adjustment

**Step 2 · Plan** Agent generates structured computation plan: `position_new = current_position * (1 + 0.15 * sentiment_score)`

**Step 3 · Compute** Symbolic layer (NumPy) executes with guaranteed floating-point precision and logged intermediate values

**Step 4 · Risk** Llama 4 Scout NIM validates new position against VaR limits via symbolic compute

**Step 5 · Execution** Mistral Medium 3.5 NIM routes order with deterministic slippage calculation

## 6.4 Continuous learning + regime-aware routing

Markets are non-stationary — models trained on 2023 data degrade measurably in 2026. Monthly LoRA retraining per agent class addresses distribution shift without full retraining costs. The FinLoRA benchmark (ICLR 2026) demonstrated that LoRA methods achieve **40.1-point average accuracy gains** over base models on financial tasks, with fine-tuning executable on 4× A5000 GPUs for under \$100 per run.

A Hidden Markov Model continuously monitors realised volatility, cross-asset correlation, and bid-ask spread dynamics to detect current market regime. Each agent class carries three regime-specific LoRA adapters:

- **Low-volatility regime** — slow mean-reversion signals, value-factor emphasis, low turnover
- **High-volatility regime** — momentum and risk-off positioning, tighter VaR limits
- **Crisis regime** — liquidity-constrained optimisation, correlation breakdown handling, elevated human-in-loop escalation

The HMM regime detector operates as a lightweight classification layer — *not an LLM* — running at sub-10ms latency. This separation ensures the regime signal itself is not subject to the temporal hallucination problems described above. NIM’s native multi-LoRA support routes each inference to the appropriate regime-specific adapter without separate model deployments or additional memory footprint.

## 7 Cost reality — NIM vs. custom GPU infrastructure

Table 5: Three-year total cost of ownership comparison

| Approach               | Annual OpEx   | CapEx (Y0) | 3-Year TCO      | Key trade-off                                                   |
|------------------------|---------------|------------|-----------------|-----------------------------------------------------------------|
| Custom GB200 cluster   | \$3.8M–\$6.4M | \$2M–\$4M  | \$13.4M–\$23.2M | Maximum performance; 7–12 MLOps engineers, 6–9 month deployment |
| Nyquist via NVIDIA NIM | \$800K–\$1.4M | \$144K     | \$2.5M–\$4.3M   | 5× optimised throughput, enterprise SLAs, fast deployment       |
| Generic cloud GPU      | \$1.2M–\$2.8M | \$0        | \$3.6M–\$8.4M   | Maximum flexibility; no optimisation, no SLAs, no domain models |

### *Nyquist annual cost breakdown — itemised*

**Tier 1 · on-prem** 16× L40S @ \$9K = **\$144K hardware** + **\$72K NIM licences**

**Tier 2 · cloud** 12× A100 reserved @ \$18.3K/yr = **\$220K** + **\$54K NIM licences**

**Meta-agent API** Claude Opus 4.7 @ ~50K calls/month = **\$120K/yr**

**Continuous learning** DGX Cloud on-demand monthly retraining = **~\$400K/yr** (36 agents × \$30K–\$70K avg per cycle)

**Observability** ClickHouse · Prometheus · Grafana · regulatory audit logs = **\$80K/yr**

**Total ~\$1.09M/year** — a 70–83% cost reduction vs. custom GB200 cluster

Break-even AUM using NIM infrastructure: **\$80M–\$120M**. Break-even AUM for custom infrastructure: **\$250M–\$400M**. This difference determines whether institutional-grade financial AI is accessible only to the top-10 hedge funds globally — or to the broader institutional market.

Additional operational advantages: **6–8 weeks to production** vs. 6–9 months for a custom cluster; 2–3 MLOps engineers required vs. 7–12 for custom operations; monthly NIM catalog updates allow adopting newly optimised models in hours vs. months of re-procurement. NIM also supports **air-gapped on-premises deployment** for regulated environments where data cannot leave the perimeter — increasingly relevant as ESMA and FCA AI governance requirements tighten.

## 8 The 13B financial foundation model — training Q3 2026

NIM provides the deployment infrastructure. The fundamental pre-training gap requires a different solution. Nyquist is training a proprietary **13B-parameter financial foundation model** scheduled for Q3 2026.

## 8.1 The BloombergGPT lesson applied

The BloombergGPT experiment (2023) trained a 50B-parameter model on 363B tokens of proprietary financial data and achieved 25-point improvements over GPT-NeoX on financial tasks. **BloombergGPT’s 50B domain-specific model outperformed GPT-3’s 175B general-purpose model on financial tasks despite having  $3.5\times$  fewer parameters.**

The inference for Nyquist’s 13B model is calibrated: domain-specific pretraining provides the largest marginal gains on tasks that actually matter for production workflows — specialised vocabulary, document structure, temporal consistency, and regulatory schema. A 13B model trained on 15TB of financial-domain data is a more credible alpha-generation substrate than a 405B model trained primarily on Common Crawl.

## 8.2 Training corpus (15TB processed)

- **30 years of SEC EDGAR filings** — 10-K, 10-Q, 8-K, proxy statements, insider transactions — 50M+ documents with temporal versioning preserved
- **25 years of earnings call transcripts** — FactSet, AlphaSense, Bloomberg archives — 2M+ transcripts with speaker diarisation and temporal indexing
- **15 years of tick data** — equities (NYSE, NASDAQ), futures (CME, ICE), options (CBOE), FX (EBS, Reuters), crypto (Binance, Coinbase) — 10TB of tick-level OHLCV + bid/ask with microsecond timestamps
- **Licensed sell-side research** — Bloomberg terminal archives, Refinitiv Eikon research notes
- **Quantitative finance academic corpus** — arXiv q-fin (20K+ papers), SSRN working papers, Journal of Finance, Review of Financial Studies back-issues

## 8.3 Architecture and why 13B

- **Base model** — Llama 4 Scout architecture (Apache 2.0 licence, modifiable for commercial deployment)
- **Parameter count** — 13B fits in a single H100 (80GB VRAM), enabling faster inference and lower latency than larger models
- **Custom tokeniser** — extended vocabulary for financial entities (CUSIP, ISIN, Bloomberg ticker formats), corporate action types, SEC filing section headers
- **Context window** — 32K tokens (sufficient for most financial documents; 1M context is not needed for the majority of quant tasks)
- **Training infrastructure** — 512× H100 GPUs on NVIDIA DGX Cloud, 6–8 week window, Megatron-LM + NeMo Framework 2.0
- **Training cost** — \$1.2M–\$1.8M for initial run

LoRA retraining cost scales linearly at 13B — monthly financial domain adaptation costs \$30K–\$70K per agent class. **At 405B, the same cycle would cost \$750K–\$1.5M per agent class**, making continuous learning economically infeasible. Smaller models also produce fewer confabulations on domain-constrained tasks, consistent with FinLoRA findings.

Post-training deployment: packaged as a NIM container, deployed across all 36 agents as base model with agent-specific LoRA adapters. A/B tested against current models on live P&L contribution over a 4-week window. Targets: **15–25% improvement in out-of-sample Sharpe ratio attribution** from signal agents; **30–40% reduction in false positive rate** from risk agents.

## 9 What actually constitutes valid evaluation for financial AI

Academic benchmarks (SWE-Bench, AIME, MMLU-Pro, ARC-AGI) are **structurally invalid** for evaluating financial AI systems. They test static knowledge recall and symbolic reasoning on closed-form problems — neither of which captures the temporal, stochastic, regime-dependent nature of financial inference. The ESMA/Turing report explicitly notes that standard AI evaluation metrics “*do not capture the specific failure modes relevant to financial applications*”.

The emerging institutional standard for 2026:

- **Scale SEAL (finance subset)** — tests document extraction, financial modelling, risk analysis. Current leader Opus 4.7 at 71%. Useful for analyst-grade AI workflows, not quant trading AI
- **FinTSB (Tongji Finance Lab, 2025)** — covers uptrends, downtrends, volatility, and black-swan event regimes; preprocessing specifically designed to prevent data leakage makes benchmark results meaningful for production evaluation
- **QuantBench 2026** — alpha generation measured by out-of-sample IC and rank correlation stability across regime shifts; portfolio construction by Sharpe, Sortino, max drawdown, Calmar; risk management by VaR breach rate, stress test scenario accuracy; zero-tolerance look-ahead bias detection

Nyquist’s per-agent production targets, measured weekly:

Table 6: Per-agent production targets

| Agent class      | Primary metric                                                | Target                |
|------------------|---------------------------------------------------------------|-----------------------|
| Data agents      | Schema validation pass rate · gap detection recall            | >99.9% · >99%         |
| Signal agents    | Out-of-sample IC · MoM rank correlation · Sharpe contribution | >0.05 · >0.80 · >1.2  |
| Risk agents      | VaR breach FP rate · stress test calibration error            | <5% · <10%            |
| Execution agents | Slippage vs. VWAP · fill rate · latency p99                   | <5bps · >98% · <600ms |

**Crisis scenario regression testing.** Every model update is tested on 2008 (Lehman collapse, correlation breakdown), 2020 COVID shock (record volatility, liquidity crisis), and 2022 rate shock (equity-bond correlation flip). A model must maintain positive risk-adjusted performance across all three before any production deployment. Human quant PM reviews 100 randomly sampled agent decisions weekly — any decision scored below 3/5 triggers model review and potential retraining.

## 10 Regulatory dimension — the hidden infrastructure requirement

The regulatory trajectory for AI in algorithmic trading adds another dimension to the infrastructure selection problem that is rarely discussed in AI capability debates.

ESMA’s February 2026 supervisory briefing on MiFID II algorithmic trading establishes specific requirements that directly constrain frontier LLM deployment:

- **Article 9 self-assessment** — firms must validate each algorithmic strategy before deployment and after any “material change”. ESMA explicitly flags that *“a series of small changes could accumulate over time into a material change in model output without it being tested”*. LoRA fine-tuning updates, by definition, trigger this requirement for every monthly retraining cycle
- **Compliance staff comprehension** — firms must ensure *“compliance staff should at least have a general understanding of how algorithmic trading systems operate”*. 60K–200K token reasoning chains are operationally incompatible with this requirement
- **Stress testing obligations** — complex algorithmic trading must be tested at *“twice the volume of the highest volume of trading reached by the firm during the previous six months”*. This mandates infrastructure that can be stress-tested on-demand, not general-purpose API endpoints

The EU AI Act further complicates frontier API deployment: AI systems *“intended to interact directly with natural persons”* in trading contexts may trigger transparency requirements. **On-premises NIM deployment under a defined governance framework provides a cleaner compliance path than opaque third-party API calls to general-purpose frontier models.**

Nyquist’s deployment design maps explicitly to these requirements:

- Deterministic symbolic compute for all calculations — auditable
- Structured per-decision logging in ClickHouse — queryable by compliance
- Human-in-loop circuit breakers with documented escalation procedures — Article 9 compliant
- On-premises deployment option — for data-perimeter requirements

## 11 Where Nyquist fits

At Nyquist, the answer to *“why even Claude Opus 4.7 fails on quant”* is structural — and the architecture that closes the gap is what we have built. **36 specialised agents on NVIDIA NIM**, each in the 13B–109B range, each with regime-specific LoRA adapters retrained monthly. **Zero arithmetic in the LLM layer** — every numerical operation routed to a symbolic engine with logged intermediate values. **A 13B financial foundation model** in training for Q3 2026 to address the pre-training gap that no general-purpose model can close. **\$1.09M/year total infrastructure cost** — institutional-grade, not hedge-fund-only.

The bottleneck has shifted from “we need bigger models” to “we need the right architecture around them.” Frontier LLMs are the reasoning and interface layer. Deterministic quantitative infrastructure is everything underneath. The split is not optional.

## Key takeaways — six sentences worth keeping

1. Frontier models hit **89–95%** on academic benchmarks (SWE-Bench, AIME, ARC-AGI) but stall below **68%** on real institutional quantitative finance — a **22–27 point structural gap**, not a scale problem.
2. Formal proofs (NeurIPS, ICML) establish that transformers with log-precision arithmetic **cannot reliably compose arithmetic** for sufficiently large instances — additional compute does not fix this.
3. Five failures persist across every May 2026 frontier model: **16–28% numerical hallucination, 35–52% PIT bias leak, catastrophic low-SNR signal extraction, <48% multi-step workflow success, 60K–200K-token reasoning traces incompatible with MiFID II audit requirements.**
4. Two Sigma, AQR, D.E. Shaw deploy LLMs as **research and interface layers, not direct alpha generators** — the industry has converged on AI-as-orchestration on top of deterministic quantitative infrastructure.
5. The solution is not bigger models — it is **36 specialised 13B–109B agents + hybrid symbolic-neural compute (zero LLM arithmetic) + a 13B financial foundation model** with continuous LoRA retraining; FinLoRA shows 40.1-point gains from domain fine-tuning.
6. **NVIDIA NIM enables production deployment at \$1.09M/year — a 70–83% cost reduction** vs. \$3.8M–\$6.4M for custom GPU infrastructure, with enterprise SLAs and 10× faster deployment.

## Constraints and limitations

- **NIM latency floor.** NIM containerisation adds 15–30ms overhead vs. bare-metal GB200 — acceptable for most quant workflows (p95 <500ms) but insufficient for ultra-low-latency HFT requiring <10ms end-to-end. Ultra-HFT remains outside the current architecture scope.
- **Performance estimate scope.** The 68% aggregate accuracy on “real quant finance tasks” reflects aggregated institutional testing — actual performance varies significantly by task type, market regime, and data quality.
- **Foundation model timeline.** Q3 2026 estimate depends on DGX Cloud cluster availability (6–10 week lead times for reserved allocation) and training convergence — both carry meaningful uncertainty.
- **Cost estimates.** Based on AWS us-east-1 reserved instance pricing and NVIDIA list pricing as of May 2026; actual costs vary by negotiated enterprise agreements and commitment terms.
- **Continuous learning effectiveness.** Monthly LoRA retraining benefits diminish in extended low-volatility regimes with minimal distribution shift — diminishing returns are a real risk in stable conditions.
- **QuantBench standardisation.** QuantBench 2026 is an emerging standard, not yet universally adopted — evaluation methodology consensus across the institutional quant community is still developing.

- **Regulatory trajectory.** ESMA’s February 2026 MiFID II briefing is non-binding guidance — actual enforcement interpretations may diverge across national competent authorities. EU AI Act interaction with algorithmic trading remains legally unsettled.
- **BloombergGPT caveat.** GPT-4 and subsequent frontier models later matched BloombergGPT on public financial benchmarks — domain-specific pretraining advantage is strongest on proprietary evaluation sets. Nyquist’s 13B model advantage will be empirically validated against live P&L, not academic benchmarks.

## Related reading

- [07 · Your backtest is lying to you — dirty market data](#)
- [06 · Quantum supremacy in finance — marketing or reality?](#)
- [05 · No, you didn’t build the next Bloomberg in a weekend](#)
- [04 · Running 36 AI agents in production](#)
- [03 · CeDeFi is not a buzzword](#)
- [02 · Why finance still doesn’t have a proper data layer](#)
- [01 · AI Agents in Trading — where the real boundary lies](#)

## References

- [1] NVIDIA. *State of AI in Financial Services 2026*. Survey of 800+ financial services professionals. NVIDIA Corporation, 2026.
- [2] Anthropic. *Claude Opus 4.7 Model Card and SWE-Bench results*. Anthropic, May 2026.
- [3] OpenAI. *GPT-5.5 Pro system card and AIME 2025 evaluation*. OpenAI, May 2026.
- [4] Google DeepMind. *Gemini 3.1 Pro ARC-AGI-2 evaluation report*. Google DeepMind, May 2026.
- [5] Scale AI. *SEAL Finance Leaderboard*. Scale AI, 2026. [scale.com/leaderboard/finance](https://scale.com/leaderboard/finance)
- [6] Ye, H., et al. *FinTSB: A Comprehensive and Practical Benchmark for Financial Time Series Forecasting*. Tongji University Finance Lab, 2025. arXiv:2501.13833.
- [7] Dong, J., et al. *QuantBench 2026: Institutional Quantitative Finance Evaluation Suite*. 2026. arXiv preprint.
- [8] Feng, Y., et al. *LiveBench: A Challenging, Contamination-Free LLM Benchmark*. 2025.
- [9] Hahn, M., & Roßin, M. *Why Transformers Cannot Compose: A Proof by Communication Complexity*. ICML 2024. arXiv:2402.09916.
- [10] Merrill, W., & Sabharwal, A. *The Expressive Power of Transformers with Chain of Thought*. ICLR 2024. arXiv:2310.07923.
- [11] Sanford, C., et al. *One-Layer Transformers Fail to Solve the Induction Heads Task*. NeurIPS 2024. arXiv:2408.14332.

- [12] Strobl, L., et al. *What Formal Languages Can Transformers Express? A Survey*. Transactions of the Association for Computational Linguistics, 2024.
- [13] Feng, G., et al. *Towards Revealing the Mystery Behind Chain of Thought: A Theoretical Perspective*. NeurIPS 2023. arXiv:2305.15408.
- [14] Abbe, E., et al. *Generalization on the Unseen, Logic Reasoning and Degree Curriculum*. ICML 2023. arXiv:2301.13166.
- [15] Wu, S., et al. *BloombergGPT: A Large Language Model for Finance*. Bloomberg LP, 2023. arXiv:2303.17564.
- [16] Lopez-Lira, A., & Tang, Y. *Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models*. SSRN 4412788, 2023.
- [17] Dong, Y., et al. *Revisiting Large Language Models as Zero-Shot Relation Extractors*. EMNLP 2023.
- [18] Xie, Q., et al. *PIXIU: A Large Language Model, Instruction Data and Evaluation Benchmark for Finance*. NeurIPS 2023. arXiv:2306.05443.
- [19] Yang, H., et al. *InvestLM: A Large Language Model for Investment Using Financial Domain Instruction Tuning*. 2023. arXiv:2309.13064.
- [20] Li, X., et al. *FinGPT: Open-Source Financial Large Language Models*. FinAI Lab, 2023. arXiv:2306.06031.
- [21] Yu, P., et al. *FinLoRA: Finetuning Quantized Financial Large Language Models Using Low-Rank Adaptation*. ICLR 2026 Workshop. arXiv:2411.18571.
- [22] Kim, G., et al. *Multi-Agent LLM Architectures for Quantitative Financial Workflows: A Benchmark*. arXiv:2603.09821, March 2026.
- [23] Chen, Z., et al. *Survey of Large Language Model Applications in Equity Markets: 84 Studies*. Frontiers in Artificial Intelligence, 2025.
- [24] ESMA. *Supervisory Briefing on Algorithmic Trading Under MiFID II*. European Securities and Markets Authority, February 2026. ESMA70-156-5654.
- [25] ESMA & The Alan Turing Institute. *Large Language Models in Finance: Capabilities, Risks and Regulatory Considerations*. Joint Report, 2025.
- [26] European Commission. *EU Artificial Intelligence Act*. Regulation (EU) 2024/1689. Official Journal of the European Union, 2024.
- [27] FCA. *AI and Machine Learning in Financial Services: Discussion Paper DP5/22 and 2025 Update*. Financial Conduct Authority, 2025.
- [28] Two Sigma. *2026 AI in Quantitative Finance Outlook*. Two Sigma Investments, Q1 2026.
- [29] D.E. Shaw Group. *AI in Systematic Investment Management: Internal Research Overview*. 2025.
- [30] AQR Capital Management. *Perspectives on Machine Learning in Asset Management*. AQR Insight Series, 2025.
- [31] NVIDIA. *Financial Services Reference Architecture: GPU-Accelerated Risk Analytics with RAPIDS*. NVIDIA Technical Report, 2026.



- [32] NVIDIA. *NIM Financial Services Reference Architecture: Multi-Agent Trading System Orchestration with NeMo*. NVIDIA, 2026.
- [33] NVIDIA. *NVIDIA AI Enterprise 5.0 Deployment Guide*. NVIDIA Corporation, 2026.
- [34] NVIDIA. *DGX Cloud: On-Demand AI Supercomputing for Enterprise*. NVIDIA Corporation, 2026.
- [35] NVIDIA. *TensorRT-LLM: Optimised LLM Inference for NVIDIA GPUs*. GitHub, 2024. [github.com/NVIDIA/TensorRT-LLM](https://github.com/NVIDIA/TensorRT-LLM)
- [36] Kwon, W., et al. *Efficient Memory Management for Large Language Model Serving with PagedAttention*. SOSP 2023. arXiv:2309.06180.
- [37] Leviathan, Y., et al. *Fast Inference from Transformers via Speculative Decoding*. ICML 2023. arXiv:2211.17192.
- [38] Sheng, Y., et al. *S-LoRA: Serving Thousands of Concurrent LoRA Adapters*. MLSys 2024. arXiv:2311.03285.
- [39] Hu, E., et al. *LoRA: Low-Rank Adaptation of Large Language Models*. ICLR 2022. arXiv:2106.09685.
- [40] Touvron, H., et al. *Llama 2: Open Foundation and Fine-Tuned Chat Models*. Meta AI, 2023. arXiv:2307.09288.
- [41] Meta AI. *Llama 4: Scout and Maverick Model Cards*. Meta AI, 2025.
- [42] DeepSeek-AI. *DeepSeek-V3 Technical Report*. DeepSeek-AI, 2025. arXiv:2412.19437.
- [43] Jiang, A., et al. *Mixtral of Experts*. Mistral AI, 2024. arXiv:2401.04088.
- [44] Qwen Team. *Qwen3 Technical Report*. Alibaba Cloud, 2025.
- [45] Wang, P., et al. *Scaling LLM Test-Time Compute: An Empirical Analysis*. 2024. arXiv:2502.06703.
- [46] OpenAI. *Learning to Reason with LLMs (o1 System Card)*. OpenAI, September 2024.
- [47] Kamath, G., et al. *Calibration of Large Language Models Using Their Generations*. 2024. arXiv:2403.14683.
- [48] Cao, Y., et al. *Risk of Data Leakage When Using LLMs as Financial Time-Series Forecasters*. 2025. arXiv:2501.16879.
- [49] López de Prado, M. *Advances in Financial Machine Learning*. Wiley, 2018.
- [50] Prado, M., & Lewis, M. *Detection of False Investment Strategies Using Unsupervised Learning Methods*. SSRN 3167017, 2019.
- [51] Bailey, D., et al. *The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality*. Journal of Portfolio Management, 2014.
- [52] Rabiner, L. *A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition*. Proceedings of the IEEE, 77(2), 1989.
- [53] Markowitz, H. *Portfolio Selection*. Journal of Finance, 7(1), 1952.
- [54] Black, F., & Scholes, M. *The Pricing of Options and Corporate Liabilities*. Journal of Political Economy, 81(3), 1973.